

Coding & Solution

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/SVNAKASH/Human-Development-Index-HDI-Prediction-Using-Machine-Learning
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	• User-friendly web interface for health and lifestyle data input. • Dataset preprocessing and cleaning. • Feature selection for machine learning. • Linear Regression model for well-being prediction. • Interactive dashboard with graphs and charts. • Prediction result display with personalized recommendations. • Model saving and loading using Joblib.
Pending / Incomplete Features	• User profile management. • Prediction history tracking. • Email notification support. • Cloud deployment and scalability improvements. • Advanced AI models for higher prediction accuracy.
Setup / Run Instructions	1. Clone or download the project. 2. Install dependencies using <code>pip install -r requirements.txt</code>. 3. Place the dataset in the project directory. 4. Run the application using <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000</code> in a web browser. 6. Enter health and lifestyle details to receive the predicted well-being score and recommendations.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application is developed using Python, Flask, and Scikit-learn to predict an individual's overall well-being based on health and lifestyle information. The system performs data preprocessing, feature selection, and Linear Regression-based prediction, then displays the results through a responsive web interface. Graphical visualizations and personalized recommendations help users understand their well-being and identify areas for improvement. The code is organized into modular components, making it easy to maintain, extend, and deploy.